

CLOUDFLARE RADAR

2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network in 2024.



17% Global Traffic Growth and Starlink's 3.3x Surge

17%

YOY TRAFFIC GROWTH

3.3x

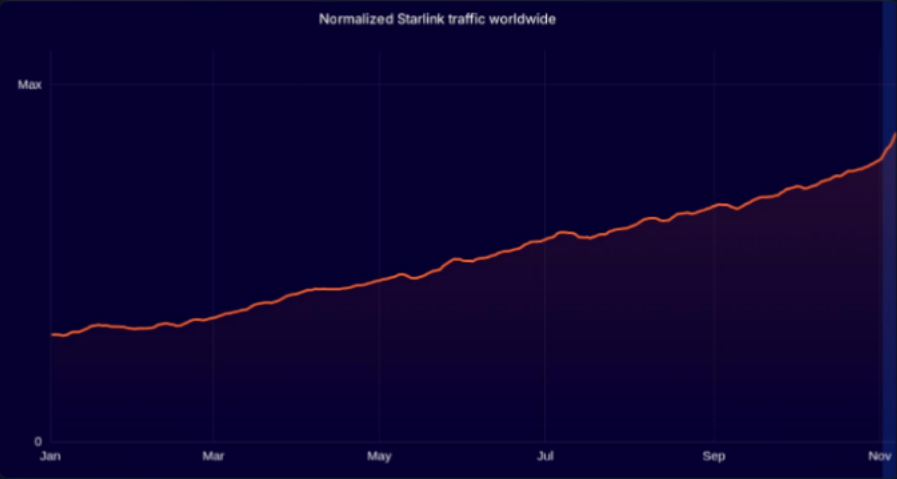
STARLINK GROWTH

Global Expansion

Global Internet traffic grew 17% in 2024, driven by increasing reliance on the Internet for communication, commerce, entertainment, and transportation.

Satellite Connectivity

SpaceX Starlink traffic saw dramatic 3.3x growth, continuing to bring satellite Internet to previously underserved locations.



Google Leads Top 10 Services as Generative AI Rises

RANK	SERVICE
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Source: Cloudflare 1.1.1.1 DNS resolver aggregate data

Rank	Service
#1	Google
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Generative AI Surge

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.



Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

POST-QUANTUM TLS 1.3

21%

HTTP/3 ADOPTION

12%

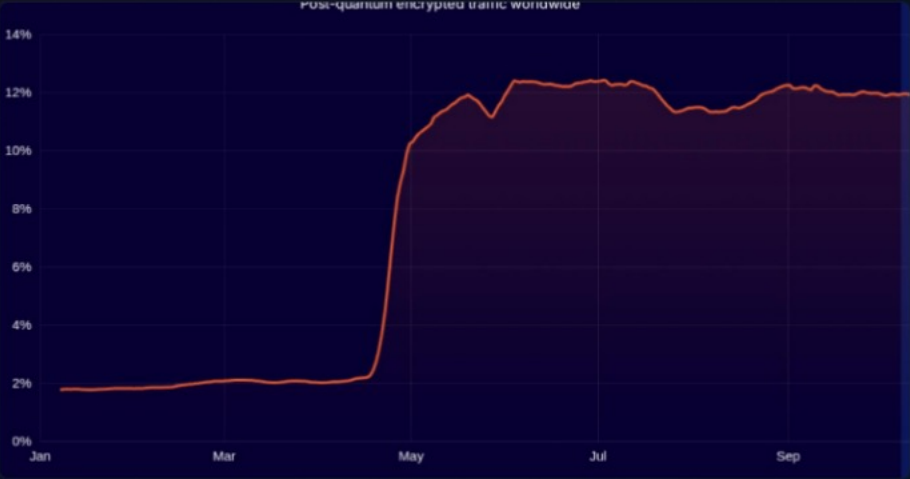
GO API CLIENT SHARE

Chrome 124 Catalyst

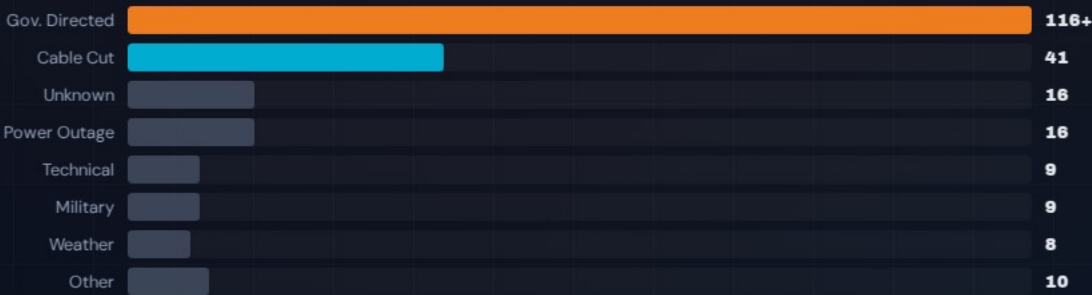
Adoption jumped sharply after Google Chrome 124 enabled post-quantum key agreement by default on April 17, 2024. Mozilla Firefox and Apple Safari have begun follow-on rollouts.

API Language Shift

Go became the #1 API client language (~12%), followed by Node.js, Python (9.6%), Java (7.4%), and .NET (3.6%).



225 Major Disruptions: Government Shutdowns Lead Outage Causes

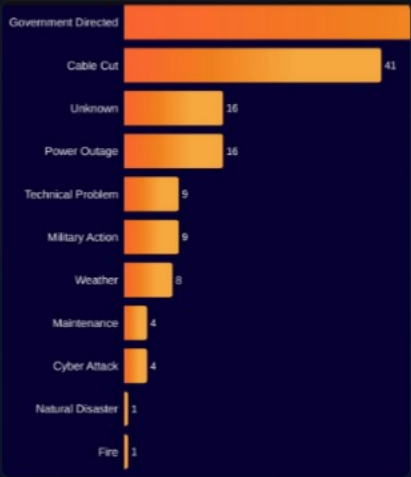


225

MAJOR DISRUPTIONS

116+

GOV. DIRECTED



Gov. directed shutdowns = 51%+ of all disruptions. Cable cuts (41) and power outages (16) followed as secondary causes.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

All traffic mitigated by Cloudflare

68%

Global bot traffic from top 10 countries

4.3%

Emails processed were malicious

43%

Malicious emails with deceptive links

Most Attacked

Gambling & Games sector was the most attacked industry category in 2024.

Log4j Persists

CVE-2021-44228 continues to be actively targeted more than two years after disclosure.

RPKI Growth

Valid routes increased 6.4%, improving Internet routing security.



Five Takeaways Shaping the 2025 Internet Outlook

- 01 Traffic Growth Trajectory** — Global Internet traffic grew 17% YoY, with Starlink surging 3.3x as satellite connectivity expands to underserved regions.
- 02 Post-Quantum Encryption Momentum** — Adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124’s default enablement; browser ecosystem is following.
- 03 Connectivity Fragility from Government Actions** — Government-directed shutdowns remained the leading cause of 225 major disruptions, followed by cable cuts (41) and power outages (16).
- 04 Persistent Security Threats** — 6.5% of traffic mitigated, Gambling/Games most targeted, and Log4j exploitation persists years after disclosure.

